Nils Blank

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Ph.D. researcher studying how spatial and temporal representations enable generalization in robot manipulation and embodied reasoning. I work on scalable supervision from robot demonstrations, pretraining for behavior generation, and training VLMs and VLAs for embodied reasoning and control.

Research Experience

Ph.D. Researcher, Karlsruhe Institute of Technology (KIT) Jul. 2024 – Present
Foundation Models for Robotics (Advisor: Prof. Rudolf Lioutikov) *Karlsruhe, Germany*

- **Ongoing research:** Investigating bidirectional attention and pretraining objectives in distilled encoder–decoder models for embodied reasoning and behavior generation, achieving state-of-the-art zero-shot robustness on LIBERO-Plus without large-scale pretraining. **Spotlight**, "Do Robots Need World Models?" Workshop, CoRL 2026. *(Under review.)*
- **SPARC (CoRL 2026):** Proposed a risk-aware spatial annotation pipeline using phase-aligned motion and gripper proximity to identify manipulated objects. SPARC-trained policies reached **31% vs. 9%** real-world success. Generated **511k VQA samples**. Models trained on this data outperformed those trained on human annotations on embodied benchmarks (**69.6% vs. 62.2%**).
- **NILS (CoRL 2024):** Developed a zero-shot pipeline that segmented and annotated **115k+ robot trajectories** (430+ hours, 7 embodiments) with natural-language instructions, removing the manual labeling bottleneck for policy learning.
- **Distributed VLM/VLA training:** Built distributed training pipelines for VLMs and robot policies on multi-node clusters (up to **64 GPUs**; SLURM, FSDP/DeepSpeed), pretraining and fine-tuning models up to **9B parameters** on **10M+** multimodal samples from self-curated robot and embodied VQA datasets.
- Set up a **ROS 2-based bimanual Franka system** for real-robot data collection and policy evaluation.

Selected Publications

SPARC: Reliable Spatial Annotations from Robot Demonstrations at Scale, CoRL 2026 *First author*
Nils Blank, Paul Mattes, Maximilian Xiling Li, Jakub Suliga, Thomas Roth, Moritz Reuss, Pankhuri Vanjani, Rudolf Lioutikov

Scaling Robot Policy Learning via Zero-Shot Labeling with Foundation Models, CoRL 2024 *First author*
Nils Blank, Moritz Reuss, Marcel Rühle, Ömer Erdinç Yağmurlu, Fabian Wenzel, Oier Mees, Rudolf Lioutikov

SIR: Structured Image Representations for Explainable Robot Learning, CVPR 2026
Paul Mattes, Jan Schwab, Jens Bosch, Maximilian Xiling Li, **Nils Blank**, Minh-Trung Tang, Moritz Haberland, Rudolf Lioutikov
Contribution: designed the real-world evaluation protocol and implemented representation sparsification.

BEAST: Efficient Tokenization of B-Splines Encoded Action Sequences for Imitation Learning, NeurIPS 2025
Hongyi Zhou et al., including **Nils Blank**, Rudolf Lioutikov
Contribution: co-developed the training framework and integrated the VLM backbone.

Education

Karlsruhe Institute of Technology (KIT) 2020 – 2024
M.Sc. in Computer Science, Study Profile: Artificial Intelligence *Karlsruhe, Germany*

- Final grade: 1.2 (German scale 1.0–4.0, 1.0 = best).
- Exchange semester at Universitat Politècnica de Catalunya, Barcelona.

Eberhard Karls University of Tübingen 2017 – 2020
B.Sc. in Media Informatics *Tübingen, Germany*

Industry Experience

Working Student, SAP Security Research Nov. 2022 – Sep. 2023
SAP Research (Advisor: Dr. Tom Ganz) *Karlsruhe, Germany*

- Researched code anonymization and privacy-preserving code generation with large language models.

Working Student, Data Science & Artificial Intelligence Mar. 2021 – Mar. 2022
1&1 IONOS SE *Karlsruhe, Germany*

- Evaluated unsupervised and deep learning methods for webspace categorization and document classification.

Technical Skills

ML / Robotics: PyTorch, VLM Training, vLLM, ROS 2, SLURM, multi-GPU / distributed training
Programming: Python, C++

Awards & Service

Best Extended Abstract Award: RSS Lab2Production Workshop (SPARC).
Spotlight: CoRL 2026 "Do Robots Need World Models?" Workshop *Robust VLAs through Cross-Modal Self-Prediction*.
Workshop Co-organizer: RSS 2026 Workshop: Trustworthy Embodied Foundation Models: Safety by Design versus Safety by Practice.
Bootcamp Co-organizer: KIT Robot Foundation Model Bootcamp, 2026; mentored 15 Ph.D. students from across Germany training robot foundation models on real hardware.
Outstanding Reviewer: LEAP Workshop, 2024.
Reviewer: CoRL, ICLR, NeurIPS, IEEE T-RO, RSS